

JUNIPERO VERBEKE

 juniperoverbeke.me  github.com/JVTou  (669) 238-8160

 linkedin.com/in/junipero-verbeke  juniperoverbeke@gmail.com

Electrical Engineering M.S. candidate with a B.S. in Applied Physics, focused on **embedded systems and device physics**, with hands-on experience in **MATLAB/Simulink, Python, QSPICE and TCAD** for research and production. Highly practical and passionate about hard tech: building and testing hardware for real environments in large-scale deployments. Experience in technical sales working with clients to implement low-voltage projects. American, French and Belgian citizen based in San Jose, CA, with mobility to U.S. and European regions. Particular interest in the Portland metropolitan area.

EDUCATION

M.S. Electrical Engineering — San Jose State University <i>GPA: 4.0</i>	Expected June 2027
B.S. Applied Physics — University of California, Santa Cruz <i>GPA: 3.5</i>	June 2025

SKILLS

Optics & Physics: Precision Optical Alignment, Fabry-Perot Interferometry, Spectroscopy, Radiation Detection, Error Analysis, Experimental Design, Fluid Dynamics, Semiconductor Physics
Hardware & Electronics: Circuit Analysis, Embedded Systems (Arduino), PID Tuning, Flight Controllers, Soldering, 3D Printing (GCODE), CNC, Data Acquisition
Software & Modeling: Python (NumPy, SciPy, Matplotlib), C, MATLAB/Simulink, Git/GitHub, LaTeX, Synopsys TCAD, Unix Shell, PSpice

EXPERIENCE

FinFET and Nanowire SRAM Radiation Hardness Studies | TCAD Simulation Framework Oct. 2025 - Present

- Extensive usage of Synopsys Sentaurus TCAD Suite (SDevice, Structure Editor, SProcess, SVisual, Sentaurus Workbench, Silicon Workbench, Sentaurus Data Explorer).
- Characterized and modeled FinFET and Gate-All-Around FET devices and SRAMs under radiation using TCAD simulation (device simulation, process simulation, layout)

Lawrence Livermore National Labs | Research Collaboration Nov. 2025 – Present

- Initiated a joint effort between LLNL & SJSU on response of a cable in a metal cavity under X-ray irradiation.
- Implemented a QSPICE environment to analyze effects of radiation-induced electromagnetic fields.

Veritas Managed Solutions, Inc. | Applied Engineer June 2021 – Present

- Spearheaded the formation of the engineering team, contributing to revenue growth from \$10M to \$28M through expansion into advanced low-voltage projects.
- Engineered and estimated over 200 bids, with large-scale system deployments at **Tesla Gigafactories NV & TX**.

Custom Mechatronic Systems Design | Personal Business June 2020 – Present

- Built, programmed, and tuned drones, integrating flight controllers, sensors, and power distribution systems.
- Designed and fabricated custom automotives components using **CAD (Fusion360), 3D printing, and CNC**.
- Performed hardware troubleshooting and soldering on electronic subsystems.

RESEARCH PROJECTS

Experimental Verification of the Zeeman Effect | Optics & Precision Alignment Jan. 2025

- Designed an optical experiment to observe the splitting of mercury spectral lines in a magnetic field, utilizing a **Fabry-Perot etalon**, CCD camera, and polarization filters.
- Performed **precision optical alignment** to project interference rings onto a sensor for data capture.
- Analyzed spectral data to calculate the Bohr magneton.

Compton Scattering & Electron Mass Determination | Instrumentation & Data Analysis Feb. 2025

- Calibrated scintillation counters, photomultipliers, and multi-channel analyzers to detect gamma-ray scattering.
- Validated the Klein-Nishina formula and Dirac theory by measuring scattered photon energy, performing **Gaussian fitting** and χ^2 error analysis on experimental data.

Physics-Based Numerical Modeling of Dynamic Systems | Python, MATLAB Nov. 2023

- Developed numerical models for dynamic systems from Lagrangian mechanics to simulate physical behaviors.
- Implemented **RK4 and Euler algorithms** in Python/NumPy to solve differential equations.

Aerodynamic Analysis & Fluid System Validation | Fluid Dynamics, XFLR5 Mar. 2025

- Investigated fluid dynamic principles using Pitot/Venturi tubes and wind tunnels; acquired sensor data (pressure, velocity) to validate Bernoulli and Navier-Stokes concepts.